

Enrollment Knowledge Transfer (EnrollmentKT)

Scope: Mobile MSC Enrollment microservice (`unite-enrollment`) — how it works, how to test manually (Postman), and how to automate (Rest Assured).

Sources scanned: `UniteMSC/unite-enrollment` , `API_Automation/api-test-automation`

Last updated: 2026-07-29

1. What enrollment is (and what it is not)

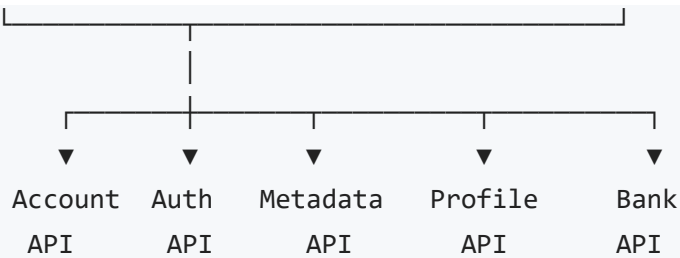
There are **three related but separate** enrollment systems in the Ascensus 529 landscape:

System	Purpose	Base URL pattern	This doc covers?
MSC Enrollment API	Mobile app step-by-step enrollment	<code>{BFF}/enrollmentapi/v1</code>	Yes — primary focus
Universal Platform (UE Web)	Browser enrollment UI + JSON account API	<code>agsup-ui-enrollment-{env}.gs.ascensus.com</code>	Referenced only
Legacy web enrollment	Plan-specific JSP flows	<code>{plan}.stage1.acs529.com/{tpl}/enrollment/...</code>	Out of scope

MSC Enrollment is an event-driven REST service. A user (prospect) moves through encrypted steps — prospect → owner → beneficiary → bank → allocations → review — and the service creates a 529 account at the end by calling downstream microservices.

2. High-level architecture





Key design points:

- Enrollment service **orchestrates**; it does not own account data long-term.
- PII is **AES-encrypted** on the client; only `prospect.plan` stays plaintext in MSC payloads.
- Flow state is chained via **event metadata** (`correlationId`, `eventId`, `seqNum`, `parent`).
- Prospect JWT (`ENROLL_PROSPECT`) is issued after successful prospect creation.

3. Codebase structure (unite-enrollment)

```
unite-enrollment/
├── src/main/java/com/cs529/enrollment/
│   ├── resource/           # REST endpoints (JAX-RS)
│   ├── service/           # Business logic
│   ├── gateway/           # HTTP clients to downstream services
│   ├── domain/            # Enrollment aggregate + related models
│   └── repository/        # Cassandra DAO, event types/topics
├── src/test/
│   ├── resources/features/ # Cucumber acceptance tests
│   └── java/enrollment/   # Step definitions
├── EnrollmentPostman/     # Postman collections, env files, run-flow.ps1
└── pom.xml                # Dependencies + integration-test profile
```

Layer responsibilities

Layer	Key classes	Role
Resource	EnrollmentResource , PlanSelectionResource , BankInstructionResource , AuthenticationResource	HTTP API, auth gates, JWT header handling
Service	EnrollmentService , OwnerService , BeneficiaryService , BankInstructionService , EnrollmentAllocationService	Validation, event production, account creation on review-confirm
Gateway	AccountGateway , AuthenticationGateway , MetadataGateway , ProfileGateway , BankGateway	Calls to account, auth, metadata, profile, bank APIs
Domain	Enrollment , Prospect , Owner , Beneficiary , Bank , EnrollmentAllocation	Request/response aggregate
Repository	EnrollmentDao , EnrollmentEventTypes , EnrollmentTopics	Cassandra persistence, event constants

4. How the enrollment flow works

4.1 Two test patterns

A. Mobile step-by-step flow (Postman / mobile app)

Each step is a separate `POST` with encrypted payload. State chains from the previous response.

B. Acceptance single-shot flow (Cucumber `enrollment.feature`)

All data sent in one `POST` to `review-confirm-entered` with a test JWT (`ACCEPTANCE_TEST` role).

4.2 Mobile flow sequence

Step	Method	Path	Auth	What happens
01	GET	/certificate	None	Client gets RSA cert; generates encAesKey
02	GET	/usstates	None	Load US states for address forms
03	GET	/plans	None	List mobile-enabled plans
04	GET	/plans/{planId}	None + x-app-version	Plan branding, metadata, fund info
05	POST	/enrollments/prospects	None	Create prospect → returns ENROLL_PROSPECT JWT
06	POST	/enrollments/enrollment/owner-entered	Bearer JWT	Owner SSN, name, DOB
07	POST	/enrollments/enrollment/owner-address-entered	Bearer JWT	Mailing/permanent address
08	POST	/enrollments/enrollment/beneficiary-entered	Bearer JWT	Beneficiary details
09	POST	/verify/routingnumber	Bearer JWT	Optional bank routing validation
10	POST	/enrollments/enrollment/bank-entered	Bearer JWT	Bank account + one-time contribution
11	POST	/enrollments/enrollment/recurring-contribution-entered	Bearer JWT	Optional recurring AIP
12	POST	/enrollments/enrollment/allocations-entered	Bearer JWT	Fund allocation percentages
13	POST	/enrollments/enrollment/review-confirm-entered	Bearer JWT	Final submit → account created

Step 00 (POST /mobile1api/v1/mobilemembersession on the **wtn** host) is for member login only. **Do not** use the member JWT for enrollment steps 06–13.

4.3 Event URL pattern

All step events (06–13) use one dynamic endpoint:

```
POST /enrollmentapi/v1/enrollments/{topic}/{event}
```

Topic	Use case
enrollment	New account enrollment
subsequentenrollment	Add account to existing member

Event (EnrollmentEventTypes)	Step
owner-entered	Owner info
owner-address-entered	Owner address
beneficiary-entered	Beneficiary
bank-entered	Bank + one-time instruction
recurring-contribution-entered	Recurring AIP
allocations-entered	Fund allocations
review-confirm-entered	Final submit

4.4 State chaining (critical for automation)

Every event after prospect creation must include a `parent` object from the **previous response**:

```
{
  "parent": {
    "correlationId": "<from prior response>",
    "eventId": "<from prior response>",
    "aggregateType": "com.cs529.enrollment.domain.Enrollment",
    "seqNum": 0
  },
  "eventId": "<new uuid>",
  "correlationId": "<same across flow>",
  "seqNum": 1,
  "eventType": "owner-entered",
  ...
}
```

Postman and `run-flow.ps1` store these in env vars: `correlationId`, `eventId`, `seqNum`, `enrollmentState`.

On `review-confirm-entered`, the service:

1. Validates all prior data via profile/bank/account gateways
2. Generates account number

3. Calls `AccountGateway.createAccount`
4. Persists snapshot to Cassandra (`EnrollmentDao`)
5. Fires system events (`enrollment-completed` , `database-sync-requested`)

5. Authentication and JWT model

Roles

Role	When issued	Can do
ENROLL_PROSPECT	After <code>POST /enrollments/prospects</code> succeeds	Steps 06–13 on topic <code>enrollment</code>
MEMBER	After mobile member login	Blocked on topic <code>enrollment</code> (401)
ACCEPTANCE_TEST	Cucumber/integration tests	All endpoints in local acceptance
VANGUARD	Partner API flow	<code>/enrollments/submit</code>
ASCENSUS_COLLABORATOR	Service-to-service	Gateway internal calls

JWT capture

After step 05 (Create Prospect), the JWT is returned in:

- Response header: `x-enc-jwttoken`
- Response body: `jwtToken` field (stripped from body on some events)

Use as: `Authorization: Bearer {{jwtToken}}`

Auth rules enforced in code (`EnrollmentResource.java`)

- Topic `enrollment` + role `MEMBER` → **401** ("You cannot access this resource")
- Topic `subsequentenrollment` → requires `ENROLL_PROSPECT` with `uii_member_id` entitlement, or authenticated `MEMBER`

Prospect session creation (behind step 05)

```
EnrollmentService.postEnrollmentProspects()  
  → AccountGateway.verifyProspects()      // POST accountapi/v1/members/verify/prospects  
  → AuthenticationGateway.createProspectSession() // POST authenticationapi/v1/prospectse  
  → JWT returned to client
```

If step 05 returns **500**, steps 06–13 will always fail with **401** (no JWT). This is a cascade failure, not broken downstream steps.

6. Encryption

Setup (step 01)

1. GET `/enrollmentapi/v1/certificate` → RSA X.509 certificate (DER, Base64)
2. Client generates AES-256 key + IV → `aesKeyPlain = base64(key);base64(iv)`
3. RSA-encrypt `aesKeyPlain` with PKCS1 padding → `encAesKey`
4. Include `encAesKey` on every enrollment payload

Field encryption rules

Field	Encrypted?
SSN, name, DOB, addresses, bank numbers, password, email, username	Yes (AES-CBC)
<code>prospect.plan</code>	No (plaintext plan branding, e.g. <code>okdirect</code>)
<code>usernameHash</code>	SHA-512(username), Base64 — not AES encrypted
<code>planId</code> on root	<code>null</code> during prospect step

Rest Assured encryption (api-test-automation)

Use `jsonapi-encryption` module:

- `MobileEncryptionHelper` — fetches cert from `GET {baseUrl}/enrollmentapi/v1/certificate`
 - `EncryptHelper` — encrypts JSON fields with `MOBILE_TYPE.ENROLLMENT`
 - CLI: `Runner -m encrypt -e stage -s enrollment -f payload.json`
-

7. Downstream service dependencies

Service	Version (pom)	Gateway	Used for
account	2.7.0	AccountGateway	Verify prospects, generate account #, create account, verify allocations
authentication	2.7.0	AuthenticationGateway	Prospect session JWT, OAuth, resume login
metadata	2.7.0	MetadataGateway	Plans, US states, countries, allocation funds
profile	2.4.0	ProfileGateway	Owner/beneficiary info and address verification
bank	2.9.0	BankGateway	Routing number verify, member banks
security	1.3.2	—	Key server (integration tests only)

Local integration tests (`mvn install -P debug`) deploy all WARs on Jetty at `https://localhost:8443` .

8. Environments

BFF hosts

Environment	Enrollment API (cloud)	Mobile login (wtn)
QC4	<code>https://unite-bff-cloud.qc4.unite529.com</code>	<code>https://unite-bff-wtn.qc4.unite529.com</code>
Stage1	<code>https://unite-bff-cloud.stage1.unite529.com</code>	<code>https://unite-bff-wtn.stage1.acs529.com</code>

Full API base: `{cloud}/enrollmentapi/v1`

Common plan branding values

Traunch	planId / branding	Example host (Stage1)
OKD	<code>okdirect</code>	<code>okd.stage1.acs529.com</code>
NJD	<code>njdirect</code>	<code>njd.stage1.acs529.com</code>
NMD	<code>nmdirect</code>	<code>nmd.stage1.acs529.com</code>
Hawaii	<code>hawaii</code>	—
Nevada Unique	<code>unique</code>	<code>ssga.stage1.acs529.com</code>

IDP Universal Enrollment (web UI — separate from MSC API)

Type	URL pattern
UI	<code>https://{plan-host}/enrollments/{branding}</code>
API	<code>https://{plan-host}/enrollments/api/stage1/plans/{branding}</code>
Centralized UE	<code>https://agsup-ui-enrollment-stage1.gs.ascensus.com/enrollments/...</code>

IDP API requires a Bearer IDP access token. MSC `enrollmentapi` does **not** use this path.

9. Manual testing with Postman

Assets (in `EnrollmentPostman/`)

File	Purpose
<code>enrollment-new.postman_collection.json</code>	JWT-aware step-by-step collection (recommended)
<code>enrollment.postman_collection.json</code>	Original collection with encryption scripts
<code>enrollment-stage1.postman_environment.json</code>	Stage1 hosts + test data defaults
<code>enrollment-qc4.postman_environment.json</code>	QC4 hosts + test data defaults
<code>run-flow.ps1</code>	PowerShell smoke runner (no Postman GUI needed)
<code>ENDPOINTS.md</code>	Endpoint reference
<code>TROUBLESHOOTING.md</code>	Failure patterns and fixes

Postman setup

1. Import `enrollment-new.postman_collection.json`
2. Import environment: `enrollment-stage1.postman_environment.json` (or `qc4`)
3. Set `planId` (e.g. `okdirect` for Oklahoma)
4. Run steps **in order** — do not skip step 01

Key environment variables

Variable	Set by	Used for
mobileMsc.host.url	Env file	Enrollment BFF (cloud host)
mobileLogin.host.url	Env file	Mobile login BFF (wtm host)
planId	Env file / manual	Plan branding
encAesKey	Step 01 script	All encrypted POST bodies
aesKeyPlain	Step 01 script	Encryption helper
jwtToken	Step 05 response	Bearer auth for steps 06–13
correlationId, eventId, seqNum	Chained from responses	Event state
enrollmentState	Chained from responses	Prior step payload reuse
jwtTokenManual	Manual paste	Bypass step 05 if you have a valid JWT

PowerShell smoke run (no Postman GUI)

```
cd unite-enrollment\EnrollmentPostman

# Stage1, Oklahoma Direct
.\run-flow.ps1 -EnvName stage1 -PlanId okdirect

# QC4, Hawaii
.\run-flow.ps1 -EnvName qc4 -PlanId hawaii

# With manual JWT (if step 05 fails but you have a token)
.\run-flow.ps1 -EnvName stage1 -PlanId okdirect -JwtTokenManual "<ENROLL_PROSPECT_JWT>"
```

Report written to: `run-report-{env}.md`

Postman rules (common mistakes)

Mistake	Symptom	Fix
Using member login JWT for steps 06–13	401	Use <code>jwtToken</code> from step 05 only
Skipping step 01	Garbled encryption / 400	Always run certificate first
Wrong host	404	Enrollment on cloud host, not wtm
Steps out of order	401 / validation errors	Run 01 → 05 → 06... sequentially
Step 05 fails (500)	Steps 06–13 all 401	Fix backend or use <code>-JwtTokenManual</code>

10. Automation with Rest Assured

Current state in `api-test-automation`

Area	Status	Location
Universal Platform enrollment (POST/PUT /accounts/)	Automated (mature)	<code>universal/jsonapi-aws-accountweb/</code>
MSC <code>enrollmentapi/v1/*</code>	Not automated (pilot only)	<code>mobile/enrollment/</code>
Encryption library	Ready	<code>jsonapi/jsonapi-encryption/</code>
Postman MSC enrollment	Manual	<code>postman/mobile/mobile-msc/</code>

Universal Platform enrollment (already automated)

These tests use schema `urn:ascensus:529:1.0:enrollment` against `aws-account-web` :

```
universal/jsonapi-aws-accountweb/src/test/java/accounts/account/  
  PostEnrollmentRequestTest.java  
  PutEnrollmentRequestTest.java  
  PutEnrollmentRequestFirstStepTest.java  
  PutEnrollmentSnapshotRequestTest.java  
  ...
```

Run:

```
mvn -f universal/jsonapi-aws-accountweb/pom.xml clean test `   
  -Pstage1-api-account-pipeline `   
  -Denvironment.properties=stage1.properties
```

Auth: OAuth2 client credentials via `ServerHttpRestApiClient` (`authentication-uri`, `server-client-id`, `server-client-secret`).

MSC enrollment automation (recommended approach)

The `mobile/enrollment/` module is a **bootstrap skeleton**. To automate MSC enrollment:

Step 1 — Base test class

Extend `BaseRequestTest` from `jsonapi-core` and call `configureMobileEncryption()`:

```
public class EnrollmentPlansTest extends BaseRequestTest {

    @BeforeClass
    public void setup() throws Exception {
        configureMobileEncryption(); // sets MOBILE_ENVIRONMENT_URL + MOBILE_STREAM=ENROLL
    }
}
```

Step 2 — Encryption before each POST

```
// Fetch cert and build encAesKey (handled by MobileEncryptionHelper)
String encryptedPayload = EncryptHelper.encryptJson(
    rawJson,
    MOBILE_TYPE.ENROLLMENT,
    environment // "stage" maps to unite-bff-cloud.stage1.unite529.com
);
```

Step 3 — Rest Assured call pattern

```
String baseUrl = getProperty("mobile-cloud-url"); // or from config

// Step 01: Certificate (no auth)
Response certResp = given()
    .baseUrl(baseUrl)
    .when().get("/enrollmentapi/v1/certificate");

// Step 04: Plan detail
Response planResp = given()
    .baseUrl(baseUrl)
    .header("x-app-version", "3.1.0")
    .when().get("/enrollmentapi/v1/plans/okdirect");

// Step 05: Create prospect (encrypted body, no Bearer yet)
Response prospectResp = given()
    .baseUrl(baseUrl)
    .header("Content-Type", "application/json")
    .body(encryptedProspectJson)
    .when().post("/enrollmentapi/v1/enrollments/prospects");
```

```
String jwt = prospectResp.getHeader("x-enc-jwttoken");

// Step 06+: Event steps (Bearer JWT required)
Response ownerResp = given()
    .baseUrl(baseUrl)
    .header("Authorization", "Bearer " + jwt)
    .header("Content-Type", "application/json")
    .body(encryptedOwnerJson)
    .when().post("/enrollmentapi/v1/enrollments/enrollment/owner-entered");
```

Step 4 — Chain state between steps

Parse each response for `correlationId`, `eventId`, `seqNum` and embed as `parent` in the next request. Mirror the logic in `EnrollmentPostman/run-flow.ps1` functions `New-EventMeta` and `Update-State`.

Step 5 — Suggested test class layout

```
mobile/enrollment/src/test/java/enrollment/
bootstrap/          # exists – framework wiring only
plans/              # GET /plans, GET /plans/{id} ← start here (read-only)
prospect/           # POST /enrollments/prospects
events/             # owner-entered through review-confirm-entered
e2e/                # full flow test
```

Config properties needed (add to `config/stage1.properties`)

```
mobile-cloud-url=https://unite-bff-cloud.stage1.unite529.com
mobile-wtn-url=https://unite-bff-wtn.stage1.acs529.com
enrollment-plan-id=okdirect
```

`enrollment-service-uri` does not exist yet in `api-test-automation` config — this is tracked as a gap.

Cucumber acceptance tests (in `unite-enrollment` itself)

For local/service-level testing without BFF:

```
cd unite-enrollment
mvn install -P debug
```

Runs Cucumber features with `ACCEPTANCE_TEST` JWT against local Jetty with seeded DB + Cassandra. Features map 1:1 to endpoints — see section 11.

11. Cucumber features map

Feature file	Endpoints tested	Stepdef
<code>enrollment.feature</code>	Full flow / review-confirm single-shot	<code>EnrollmentsStepdefs</code>
<code>owner.feature</code>	<code>owner-entered</code> , <code>owner-address-entered</code>	<code>OwnerStepdefs</code>
<code>beneficiaries.feature</code>	<code>beneficiary-entered</code>	<code>BeneficiaryStepdefs</code>
<code>bankinstruction.feature</code>	<code>bank-entered</code> , <code>recurring-contribution-entered</code>	<code>BankInstructionStepdefs</code>
<code>enrollmentallocation.feature</code>	<code>allocations-entered</code>	<code>EnrollmentAllocationStepdefs</code>
<code>planselection.feature</code>	<code>GET /plans</code> , <code>GET /plans/{id}</code>	<code>PlanSelectionStepdefs</code>
<code>usstates.feature</code>	<code>GET /usstates</code>	<code>UsStatesStepdefs</code>
<code>subsequenterollment.feature</code>	<code>subsequenterollment/review-confirm-entered</code>	<code>SubSequentEnrollmentStepdefs</code>
<code>vanguard.feature</code>	<code>POST /enrollments/submit</code>	<code>VanguardStepdefs</code>
<code>contentservice.feature</code>	<code>GET /content</code>	<code>ContentServiceStepdefs</code>
<code>enrollmentallocationfunds.feature</code>	<code>POST /enrollmentallocationfunds/get</code>	<code>EnrollmentAllocationFundsStepdefs</code>
<code>upromiseaccountexchange.feature</code>	<code>GET /upromiseaccount</code>	<code>UpromiseAccountExchangeStepdefs</code>

12. Supporting endpoints reference

Method	Path	Auth	Purpose
GET	/certificate	None	RSA cert for encryption setup
GET	/usstates	None	US states list
GET	/country	None	Countries list
GET	/plans	None	Mobile-enabled plans
GET	/plans/{id}	None + x-app-version	Plan detail (426 if app version too old)
POST	/enrollments/prospects	None	Create prospect + issue JWT
POST	/enrollments/{topic}/{event}	ENROLL_PROSPECT JWT	All enrollment step events
POST	/enrollments/enrollmentstarted	None	Mark enrollment started
POST	/verify/routingnumber	ENROLL_PROSPECT	Validate bank routing number
POST	/enrollmentallocationfunds/get	None	Age-based portfolio lookup
GET	/content?id=&lang=&name=	ENROLL_PROSPECT	CMS content by branding
GET	/subsequenteenrollment/banks?enAesKey=	ENROLL_PROSPECT / MEMBER	Banks for add-account flow
POST	/enrollments/submit	VANGUARD	Partner single-shot API enrollment
POST	/oauth/token	None	Partner OAuth client credentials
GET	/upromiseaccount	None	Upromise referral lookup
GET	/ping , /health/*	None	Health checks

13. Common failures and how to diagnose

Symptom	Likely cause	Action
Step 05 → 500	<code>AccountGateway.verifyProspects</code> failing behind BFF	Escalate with <code>requestId</code> from response body; check account API health
Steps 06–13 → 401 after 05 failed	No <code>ENROLL_PROSPECT</code> JWT issued	Fix step 05 first, or use <code>-JwtTokenManual</code>
Step 06–13 → 401 with valid JWT	Using <code>MEMBER</code> JWT instead of <code>ENROLL_PROSPECT</code>	Re-run step 05; check JWT role
404 on <code>/enrollmentapi</code>	Wrong host (wtm instead of cloud)	Use <code>mobileMsc.host.url</code> / cloud BFF
426 on <code>/plans/{id}</code>	<code>x-app-version</code> header missing or too old	Set <code>x-app-version: 3.1.0</code> or higher
Encrypted fields rejected	Skipped step 01 or wrong <code>encAesKey</code>	Re-run certificate step
IDP API → <code>Access token not found</code>	Missing Bearer IDP token	Separate from MSC flow; get IDP token first

Stage1 observed failure (okdirect, 2026-07-28): step 05 returned 500 (`requestId: d99599c3-5fc0-911e-a366-8514dc019eb7`). Plan endpoint (`GET /plans/okdirect`) returned 200 with `mobileEnabled=1` — issue is in prospect verification backend, not plan config.

14. Automation coverage summary

Flow	Manual (Postman)	Rest Assured (api-test-automation)	Cucumber (unite-enrollment local)
GET plans / plan detail	Yes	Not yet (recommended first target)	Yes (planselection.feature)
Create prospect	Yes (run-flow.ps1)	Not yet	Yes (enrollment.feature)
Owner / beneficiary / bank steps	Yes	Not yet	Yes (per-feature files)
Review-confirm (account creation)	Yes	Not yet	Yes (enrollment.feature)
Subsequent enrollment	Yes	Not yet	Yes (subsequenteenrollment.feature)
Vanguard API submit	Yes	Not yet	Yes (vanguard.feature)
Universal Platform /accounts/	Postman (universal-platform)	Yes (12+ test classes)	N/A

Recommended automation roadmap:

1. GET /enrollmentapi/v1/plans/{planId} — read-only, lowest risk
2. POST /enrollments/prospects — requires encryption + backend health
3. Full E2E event chain — mirror run-flow.ps1 in Rest Assured

15. Quick reference commands

Postman / PowerShell

```
# Full MSC enrollment smoke (Stage1, Oklahoma)
.\unite-enrollment\EnrollmentPostman\run-flow.ps1 -EnvName stage1 -PlanId okdirect

# Cucumber acceptance (local, all services)
cd unite-enrollment && mvn install -P debug
```

Rest Assured (Universal Platform — working today)

```
mvn -f api-test-automation\universal\jsonapi-aws-accountweb\pom.xml clean test `
-Pstage1-api-account-pipeline -Denvironment.properties=stage1.properties
```

Encryption CLI (payload prep)

```
cd api-test-automation\jsonapi\jsonapi-encryption
java -cp ... core.encryption.runner.Runner -m encrypt -e stage -s enrollment -f payload.js
```

16. Key files index

Need to understand...	Read this file
Endpoint list + flow order	EnrollmentPostman/ENDPOINTS.md
Failure patterns	EnrollmentPostman/TROUBLESHOOTING.md
Automated smoke runner	EnrollmentPostman/run-flow.ps1
Stage1 env vars	EnrollmentPostman/enrollment-stage1.postman_environment.json
Auth rules	src/main/java/.../resource/EnrollmentResource.java
Event types	src/main/java/.../repository/EnrollmentEventTypes.java
Core business logic	src/main/java/.../service/EnrollmentService.java
Account service calls	src/main/java/.../gateway/AccountGateway.java
Prospect JWT creation	src/main/java/.../gateway/AuthenticationGateway.java
Cucumber happy path	src/test/resources/features/enrollment.feature
Rest Assured encryption	api-test-automation/jsonapi/jsonapi-encryption/.../MobileEncryptionHelper.java
MSC automation pilot	api-test-automation/mobile/enrollment/README.md
Universal enrollment tests	api-test-automation/universal/jsonapi-aws-accountweb/.../PostEnrollmentRequestTest.java